VISTA—Vision-Integrated Smart Traffic Automation

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***Abstract*—Traffic Signal Control (TSC) plays an increasingly important key role in transportation in cities, where efficiency, safety, and fairness in using road infrastructure must be ensured in a dynamically changing traffic environment. Traditional traffic control systems, such as fixed-time control, sensor control, and reinforcement learning control, are inefficient in reacting to real-time congestion, emergency, and violation of traffic regulations. In this paper, an Artificial Intelligence (AI) smart traffic automation system has been designed. The proposed framework uses a combination of computer vision and decision making through language to achieve traffic assessment and control in real time. The framework uses the YOLO perception module to analyze live video feeds to estimate the number of vehicles, pedestrians, ambulances, and traffic congestion for dynamic traffic signals. The framework uses the Large Language Model-driven approach as the intelligent decision controller. The framework uses a meta-controller designed in safety-related meta-control to pick the reasoning type for emergencies and the use of reinforcement controls for regular traffic flow. The framework then uses a planning agent to calculate optimal coordinated signals based on traffic density and priorities. With the integration of perception, reasoning, and control capabilities, the proposed solution provides a scalable and intelligent system for traffic management in the contemporary world.**

***Index Terms*—YOLO, Computer Vision, Reinforcement Learning(PPO), Large Language Models (LLMs), Perception Agent, Meta Controller, SUMO Simulation.**

I. Introduction

India is going through a phase of rapid urbanization, with cities expanding quickly in terms of population and infrastructure, which has, in turn, led to an increased number of vehicles on Indian roads. The combination of these factors has led to significant traffic congestion at intersection points in Indian cities. Waiting times, fuel consumption, air emissions, and commuter irritation are some of the prevailing problems faced. Indian road traffic is further made worse because of the presence of cars, two-wheelers, pedestrians, and ambulances operating on the road in a way that fails lane discipline.

The conventional system in traffic signaling uses fixed intervals of time, usually measured in seconds time settings which are not adaptable to the actual conditions on the road. The system does not take into consideration factors like road traffic density, congestion, or emergency conditions, hence resulting in inefficient traffic flow. Even the latest systems that utilize sensors or have basic traffic density measurement capabilities still lack the intelligence to take context-specific and adaptive measures. There is, thus, an immense need for an intelligent traffic control system.

This project discusses the AI-Powered Smart Traffic Automation System, which has been conceptualized to overcome current limitations with the aid of adaptive and intelligent control over traffic signals. The system has the functionality to continuously analyze traffic patterns with the help of live camera images fed into the system, thus applying artificial intelligence to understand traffic flow patterns, which helps to control traffic signals to optimize traffic flow rather than operating within predetermined cycles.

This proposed system combines three essential elements, which are perception, decision-making, and planning functionality. The perception component is responsible for analyzing real-time images of vehicles, whereas the decision component assesses regular and critical events like the passage of emergency vehicles, and finally, the planning component calculates optimal control of signals for smoother flow of traffic. Thus, through the integration of intelligent observation and adaptive control, this system is meant to alleviate congestion and enhance safety and efficiency in urban transport systems.

II. Related Work

Within this section, we examine the literature to understand what has been developed for intelligent traffic control systems, specifically with regards to traffic signal control, emergency

vehicle handling, pedestrian identification, and traffic violation algorithms.

There have been a few studies in the area of intelligent traffic management solutions via Reinforcement Learning techniques, Computer Vision, etc. CNN, LSTM, A2C-based congestion management was introduced in the work of Ma et al. [1], but the limitations exist in the event of severe congestion, and the collaboration does not guarantee good temporal generalization. Vision-based density management via YOLO was explored in the work of Ubaid et al. [3], but the effect of occlusions was observed in the case of dense traffic flow. For better collaboration, a Multi-Agent Transfer Reinforcement Learning was introduced in the work of Ge et al. [4].

Traffic violation detection has mainly concentrated on vision-based approaches. YOLOv3-based models have been developed by Ravish et al. [5] and Malik et al. [6] for the detection of red light safety violators, but these models are not fully competent. Reddy et al. [7] and Kavitha et al. [8] developed models based on the R-CNN approach; these models were not executed in real-time with reduced complexity

Prioritization related to emergency vehicles has also been dealt with through dynamic traffic signal control algorithms. A system named 'Life Saver' was designed specifically for ambulance priority by Rajeshwari et al. [9]. Combining the advantages of cloud processing and reinforcement learning with YOLO algorithms was discussed in the works of Deoghare et al. [10] and Rajesh et al. [11] but with scalability and network dependency limitations. A PLC system based on ambulance identification has also been designed by Nalini et al. [12] but with high cost and complex control constraints.

There have also been studies that focus on pedestrian-aware traffic signal control. Real-time pedestrian and vehicle detection was used in the optimization of traffic signals by Gomathi et al. [13] in their approach, which did not consider the event of emergencies. LiDAR-based pedestrian detection through a sensor system was analyzed by Sprute et al. in their paper [14]; the problem of occlusion was tackled by a feature fusion-based approach by Karthika et al. in their paper [15], but it is not suited for real-time processing.

III. Proposed System Architecture

The proposed system solves the problem of rising traffic congestion and inefficient traffic signals. It concentrates on real-time traffic understanding and adaptive traffic signal optimization. It makes use of real-time video feeds for analysis, as well as historical traffic data for controlling the system. Further below will be a brief description of the system architecture.

The overall architecture is made up of three integrating parts: Traffic Perception, Intelligent Decision-Making, and Adaptive Signal Planning.

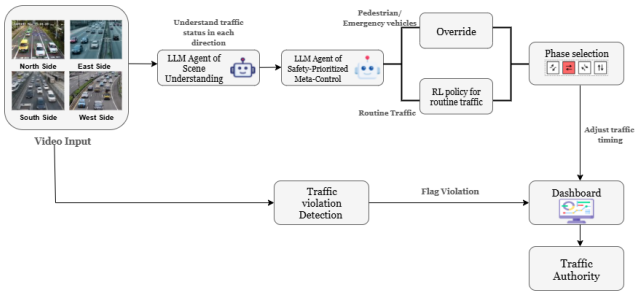

Fig. 1: Proposed System Architecture for VISTA

A. Traffic Perception Module

The Traffic Perception Module is actually the base layer of the proposed architecture. The key task of this module is to extract real-time traffic information from the physical world and convert it to formatted data that can be processed by other advanced modules. In this case, this module makes use of videos shot from monitoring cameras that are strategically located at intersections to cover all entries and exits.

The video streams are then processed by using a YOLO-based deep learning technique for object detection. YOLO is chosen because it has high accuracy rates for object detection, along with low latency, making it appropriate for a real-time system. The proposed system is trained to detect multiple classes of objects such as cars, buses, trucks, two-wheelers, pedestrians, and emergency vehicles. Multiple class object detection facilitates the proposal system in dealing with diverse traffic compositions, which are encountered in urban areas.

After the objects have been detected in the image, the detected ones Vehicles are then mapped into regions corresponding to the using geometric transformations and ROI to predefined lanes. masking. This has made counting of vehicles lane by lane possible while performing traffic density computation at each lane.

Apart from car density estimation, the perception module is also responsible for observing pedestrian existence around crosswalks and detecting emergency vehicles using visual characteristics such as sirens, color patterns, and priority signs. The identification of emergency vehicles is viewed as an critical event; hence, it affects decision-making in other modules.

The output of Traffic Perception Module includes lane-wise traffic densities, pedestrian presence, and emergency vehicles. These outputs combined form the traffic state and act as an input for Intelligent Decision-Making Module.

B. Intelligent Decision-Making Module

The Intelligent Decision Making Module is the central module of the proposed traffic control system. This module is constantly analyzing traffic patterns in real-time and providing optimal decisions on traffic signal control. This module is controlled by an intelligent controller that uses rule-based

reasoning, reinforcement learning, and decision support by large language models.

At the decision point, the system considers the level of lane congestion, pedestrian demand, and the availability of emergency cars. A meta-controller with the focus of ensuring road safety initially assesses the road traffic and chooses the control strategy based on the level of priority.

When the presence of an emergency vehicle is recognized, the system quickly initiates a reasoning mode with the objective of giving priority to the movement of emergency vehicles through the intersection.

Under typical traffic patterns, the system uses a reinforcement learning algorithm to perform a reinforcement learning-based control strategy in an optimal manner. Based on past and real-time traffic patterns, the system adjusts to different traffic patterns, congested hours, and unforeseen disturbances to achieve optimal results.

The incorporation of reasoning capabilities within the LLM increases the adaptability and explainability of the model as it considers additional knowledge regarding traffic rules, regulations of safety, and priority handling. This allows the controller to produce credible and authentic decisions even under challenging traffic conditions.

C. Adaptive Signal Planning Module

The Adaptive Signal Planning Module is responsible for converting high-level control decisions into traffic signal patterns that can be implemented. This module computes adaptive signal timings for each phase based on information provided by the Intelligent Decision-Making Module. In contrast to existing fixed-cycle methods, the proposed system adjusts signal cycle times based on real-time traffic patterns.

For each signal phase i, the green time allocation is computed as:

$$T_i = T_{\text{min}} + \alpha_i$$

where T_{min} is a minimum possible green time value needed to be ensured, and α_i is a scale factor which is calculated using the corresponding lane density value, queue length, and priority weight. The scale factor allows to distribute green time proportionally to congested lanes with high traffic density.

In the case of emergencies, the system overrules the standard timing of the signals, providing immediate right of way to the phase involved in the situation. This kind of preemptive system of control optimizes response rates for fire vehicles, ensuring safety on the roads.

The Adaptive Signal Planning Module is also mindful of other constraints like the limitations of the green time duration, crossing requirement of pedestrians, as well as the inter-green intervals. This ensures that the adaptation is done within the guidelines of the traffic rules. The signal planner changes the strategies of the timings depending upon the feedback regarding the perception and decision-making modules present in the control cycles.

D. System Integration and Workflow

The proposed system would function as a highly integrated analyzed by the Traffic Perception Module, and meaningful traffic features are extracted. These features are further analyzed by the Intelligent Decision Making Module, and optimal control strategies can be obtained. These strategies are further is accomplished by the Adaptive Signal Planning Module. This This iterative process of feedback is what enables the system to learn from what it does. Based on its perception, reasoning, and adaptive plans, the proposed system can provide efficient traffic management, relieve congestion on roads, improve safety, and be very responsive.

On the whole, the proposed Smart Traffic Automation Framework using AI provides a scalable and intelligent solution approach towards designing the next-generation traffic control solution. It is a key element in future smart city transport infrastructure.

E. Car Accident Detection System

The proposed framework includes the implementation of a Car Accident Detection System to recognize road accident occurrences at road junctions through live video streams. The system increases road safety by detecting instances of road accidents and unconventional interactions of vehicles to take corrective action through the intelligent road infrastructure.

The video streams recorded from cameras are processed on a frame-by-frame basis by the YOLO-based deep learning algorithm that has been trained using accident-related features from videos. For the input video recorded at time t, we shall denote it by I_t. The accident prediction on every video frame is performed using the detection function as follows:

$$A_t = f(I_t) \tag{1}$$

where $f(I_t)$ denotes the learned model for accident detection and A_t symbolizes the expected accident value for the corresponding frame.

For effective detection, a confidence decision mechanism is employed. An accident can be declared only when the confidence value, C_t, crosses a certain threshold, T_a. The decision-making process is expressed as:

$$D_t = \begin{cases} 1, & \text{if } C_t \geq T_a \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $D_t = 1$ is the appearance of an accident.

When the detection is successful, the system annotates the corresponding video frame by drawing bounding boxes around the accident region and showing an alert message along with the percentage confidence. The confidence value shown on the frame is computed as:

$$\text{Confidence}(\%) = C_t \times 100 \tag{3}$$

These detected accident events are passed to the Intelligent Decision Making Module, enabling the system to start the

process of priority handling operations such as the preempting of emergency signals and generation of warnings. With the integration of the accident detection system with the perception layer based on vision, road safety is enhanced.

F. Traffic Signal Violation Detection System

The Traffic Signal Violation Detection System covers a series of intelligent algorithms to automatically detect the running vehicles violating red traffic signals at an urban intersection. This subsystem interacts with the traffic perception and adaptive signal planning modules in order to increase the traffic discipline and intersection safety.

For every incoming video frame I_t, a YOLO-based object detection model identifies the relevant traffic entities, which include vehicles and traffic lights. The object detection is mathematically represented as:

$$O_t = f(I_t) \tag{4}$$

where $f(I_t)$ denotes the object detection function and O_t holds a list of identified objects in image t.

In order to calculate the state of the traffic signal, the system evaluates the predefined red-light area and extracts its average brightness value, B_r. The red signal is said to be active if its brightness value exceeds the predefined threshold value, T_b, as follows:

$$S_r = \begin{cases} 1, & \text{if } B_r \geq T_b \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

Violation of traffic signal: The violation of the traffic signal occurs when a car, after crossing the Stop Line, enters the Region of Interest (ROI) when the traffic signal is red. The violation expression can be given as:

$$V_t = \begin{cases} 1, & \text{if } S_r = 1 \wedge P_o \in ROI \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where P_o denotes the bounding box coordinates of the detected vehicle.

For enhanced reliability of vehicle detections, only those detections that have a confidence level of threshold value $\mathrm{T_c}$ or higher would be treated as actual detections. Once a violation has been detected, the system visually marks the violator vehicle, the region of interest, as well as generates a warning message accompanied by a confidence level value.

Through the integration of automated traffic signal violation detection in the proposed system, the proposed system encourages safe driving practices and supplements adaptive signal control to manage traffic in an urban area in an efficient manner.

IV. Result And Analysis

This section presents the performance evaluation of the proposed Intelligent Traffic Signal Control System using simulation-based experiments. The aim is to evaluate the efficiency of the system in optimizing the traffic flow, reducing the queue length of the vehicles, minimizing travel time, and handling the prioritization of emergency vehicles efficiently. To this end, a comparison with fixed-time signal control and an adaptive control approach using a Deep Q-Network is presented to highlight the merits of the proposed PPO-based model under dynamic conditions in traffic flow.

A. Comparison of Policy Performance

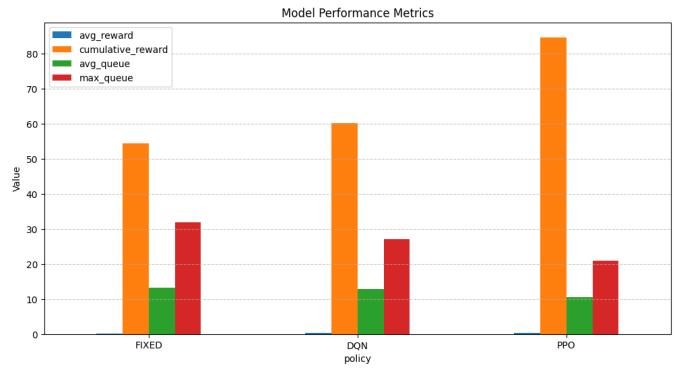

Fig. 2: Comparison of Policies Based on Reward and Queue Metrics

Fig.2 shows the comparison in terms of reward and queue-related metrics for Fixed-Time, DQN, and PPO control policies. The PPO model yields the best average reward with a value of 0.338, while DQN and Fixed-Time control report an average reward of 0.241 and 0.218, respectively, which shows that the PPO has learned a more effective policy than DQN.

With respect to cumulative reward, the PPO reports a value of 84.71, compared with 60.19 for DQN and 54.44 for Fixed-Time control, indicating consistent long-term optimization. The PPO-based approach reduces congestion more.

The queue length is minimized to an average of 10.55 vehicles, while for DQN and Fixed-Time approaches, this is recorded at 12.85 and 13.23 vehicles, respectively. Besides this, PPO has limited the maximum queue length to 21 vehicles, against the 27 for DQN and 32 for Fixed-Time control.

B. Queue Length and Trip Time Analysis

Fig.3 compares the performance of the three traffic control strategies by average queue length and normalized trip time. The Fixed-Time strategy is used as the baseline, whose normalized trip time is set as 100%. The DQN-based model performs moderately well by reducing the trip time to 97.15%, but its adaptability is still poor in the case of highly variable traffic.

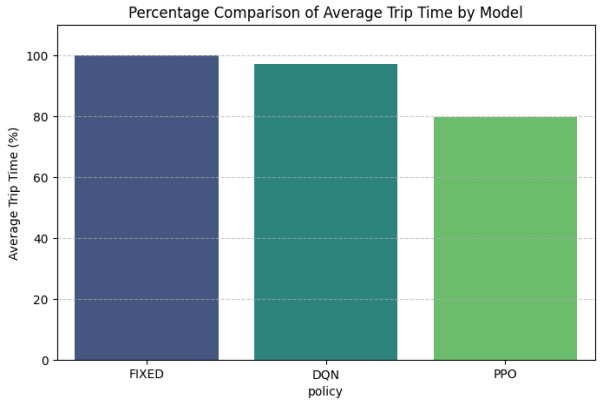

Fig. 3: Comparative performance models

In contrast, the PPO-based model achieves the lowest average queue length of 10.55 vehicles and reduces the trip time percentage to 79.74% that corresponds to nearly a 20% reduction of travel time. In such ways, this improvement demonstrates PPO's capability of adapting fast to the change in traffic density and promoting smoother flow of vehicles, leading to higher throughput at the intersection.

These experimental results confirm that the proposed PPO-based adaptive control strategy outperforms both Fixed-Time and DQN-based methods in terms of all the compared metrics, while maintaining robustness and adaptability with more efficient learning for real-time intelligent traffic signal control within dynamic urban environments

C. Performance Evaluation of the Ambulance Detection Model

The performance metrics of YOLOv11-based ambulance detection system are presented in Table I. The system offers a precision level of 97.58%, which indicates a low false-positive rate and thus proper identification of ambulances. The system's recall value of 94.50% indicates a successful detection of most ambulances available in a traffic scene, which is quite critical in emergency situations and managing traffic.

Metric	Value
Precision (P)	97.58%
Recall (R)	94.50%
mAP@0.50	99.01%
mAP@0.50:0.95	85.59%
Fitness Score	85.59%

Table I: Performance Metrics

As evident in Table I, the model achieves a mAP@0.50 of 99.01%, which shows near-perfect accuracy. In addition to this, a high mAP@0.50:0.95 of 85.59% shows competent results. In summary, it can be effectively stated that this paper asserts through results that the technique developed is quite competent for real-time ambulance detection in an intelligent transportation system.

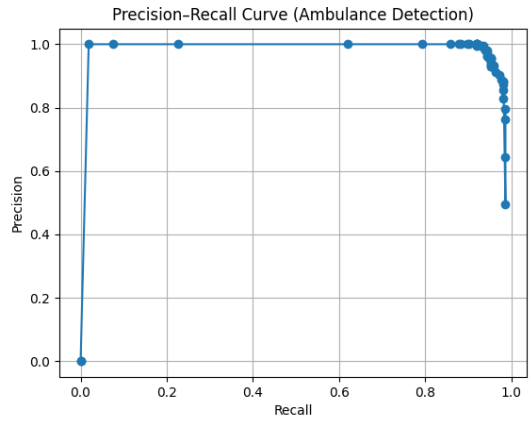

Fig. 4: Precision–Recall Curve for Ambulance detection model

Fig.4 presents the Precision–Recall curve for the proposed YOLOv11-based ambulance detection model. The curve reflects that precision remains high over a wide range of recall values; that is, while the model manages to detect most instances of ambulances, the number of false positives remains pretty low. Precision remains close to unity until the recall approaches its maximum, reflecting the strength of the model in detecting ambulances at different confidence thresholds.

D. Performance Evaluation of the Road Accident Detection Model

Table II presents the performance metrics of the proposed YOLOv11-based road accident detection system. It is evident that the precision value of 88.05% obtained is optimal for distinguishing between accident and non-accident scenarios of traffic, and this is achieved with a controlled rate of false positives. The value of 77.94% for recall highlights the efficiency of the system in detecting a large number of accident scenarios, which is important for immediate actions during traffic observation.

Metric	Value
Precision (P)	88.05%
Recall (R)	77.94%
mAP@0.50	88.76%
mAP@0.50:0.95	44.99%
Fitness Score	44.99%

Table II: Performance Metrics

As is evident from Table II, it is recorded that the mAP@0.50 value is 86.76%, thus ensuring robust detection accuracy. Additionally, with a value of 44.99% for mAP@0.50:0.95, it is clear that the localization accuracy is fairly good when measured on tougher parameters. This could be attributed to the varying nature and complexity of road accidents. From this experiment and discussion, it is clear that the proposed solution is effective for road accident detection in automated traffic analysis.

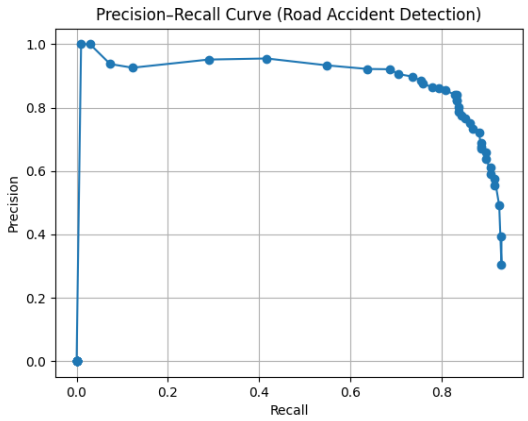

Fig. 5: Precision–Recall Curve for Accident detection model

Fig.5 shows the Precision–Recall curve for the YOLOv11-based road accident detection model. It reflects high precision over a wide range of moderate recall values but decreases gradually with increased recall. This is indicative of an evident trade-off between the coverage of detected scenes and false positives, due to the diverse visual nature and complexity of road accident scenes. In spite of such a trade-off, the model reflects very good general detection performance.

V. Conclusion

The proposed system improves the efficiency of traffic management in urban settings in India through computer vision and reinforcement learning. The method applies vehicle detection utilizing YOLO and PPO-based reinforcement learning for dynamic changes in traffic signal phases. In order to address traffic congestion and emergencies in an urban setting, there is a need for improvement in the proposed system and its application to handle complex scenarios.

Going beyond single intersection control, even more sophisticated applications of this system can be accomplished for multi-junction coordination, V2I communication, and congestion prediction modeling. Furthermore, to increase its effectiveness, In the management of emergencies and general traffic flow, more attention needs to be paid sophisticated methods of sensing and communication may be envisaged. On the matter of conceptual extension, this system will assist in amalgamating overall traffic analysis for cities as well as realtime decision-making for faster, safer, and more environmentally friendly transport solutions for cities. This work is an addition to developing an intelligent system for traffic management for smarter cities and upgraded passenger experience in highly complex urban areas.

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